🚀 Full-Stack Project Roadmap (WayStay)

Audience: 2nd-year student building a resume project
Commitment: 10–15 hrs/week
Tech Stack: Spring Boot (backend), MySQL (database), React + Tailwind (frontend), JWT (auth)

🎯 Project Goal

Deliver a **working, deployed web app** that demonstrates:

- Secure REST API design
- Database integration
- Authentication & authorization with JWT
- CRUD + relationships (users, hotels, bookings, blogs)
- Frontend with protected routes and state management
- Deployment + documentation

🧩 Core Features

1. User Authentication (Foundation)

Backend (Spring Boot):

- `User` entity: `id`, `name`, `email`, `password` (hashed with Bcrypt).
- Endpoints:
  - `POST /api/auth/register` → register user.
  - `POST /api/auth/login` → return JWT.
  - `GET /api/users/me` → return logged-in user's profile.
- Spring Security + JWT filter for request authentication.

Frontend (React):

- Pages: `/register`, `/login`.
- Store JWT in `localStorage`.
- React Context for global auth state.
- Protected routes (redirect unauthenticated users).

- Profile page: display data from `/api/users/me`.

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2. Hotels & Bookings (Core Logic)

Backend (Spring Boot):

- `Hotel` entity: `id`, `name`, `location`, `description`, `price_per_night`, `image_url`.
- `Booking` entity: `id`, `check_in_date`, `check_out_date`, linked to `User` and `Hotel`.
- Endpoints:
  - `GET /api/hotels` → list hotels.
  - `GET /api/hotels/{id}` → hotel details.
  - `POST /api/bookings` (protected) → create booking.
  - `GET /api/my-bookings` (protected) → user's bookings.

Frontend (React):

- Pages:
  - `/` (home) → list hotels (grid of HotelCards).
  - `/hotels/{id}` → hotel detail + booking form.
  - `/my-bookings` → user's bookings.
- Components: `HotelCard`, `BookingForm`.

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3. Blog (CRUD Showcase)

Backend (Spring Boot):

- `Blog` entity: `id`, `title`, `content`, `created_at`, linked to `User`.
- Endpoints:
  - `GET /api/blogs` → list posts.
  - `POST /api/blogs` (protected) → create post.
  - `PUT /api/blogs/{id}` (protected) → edit post (author only).
  - `DELETE /api/blogs/{id}` (protected) → delete post (author only).

Frontend (React):

- Pages:
  - `/blog` → list all blogs.
  - `/blog/new` → create post (protected).
  - `/blog/edit/{id}` → edit post (protected, author only).

🗁 Overall Project Structure

Root Directory

```
waystay/
├── backend/                  # Spring Boot project
│   ├── src/main/java/com/waystay/
│   │   ├── controller/       # REST Controllers
│   │   ├── model/            # Entities (User, Hotel, Booking, Blog)
│   │   ├── repository/       # Spring Data JPA Repositories
│   │   ├── security/         # JWT, filters, config
│   │   └── service/          # Business logic
│   ├── src/main/resources/
│   │   └── application.properties  # DB config, JWT secret
│   └── pom.xml               # Maven dependencies
│
├── frontend/                 # React project
│   ├── src/
│   │   ├── components/       # Reusable UI components (HotelCard, etc.)
│   │   ├── context/          # AuthContext for state management
│   │   ├── pages/            # Register, Login, Home, Hotels, Bookings, Blog
│   │   ├── services/         # API calls (authService.js, hotelService.js,
etc.)
│   │   └── App.jsx           # Routing + Protected Routes
│   ├── public/
│   └── package.json
│
├── README.md                 # Project documentation
├── .gitignore
```

📅 10-Week Student Roadmap

Phase 1: Setup & Auth (Weeks 1–3)

Week 1:

- Create GitHub repo.
- Setup Spring Boot backend + connect to MySQL.
- Setup React frontend with Vite/CRA + Tailwind.

Week 2:

- Implement `User` entity + Spring Security.
- Build `/register`, `/login`, `/users/me`.
- Test with Postman.

Week 3:

- Frontend login/register forms.
- JWT handling + Auth Context.
- Protected routes + Profile page.
  ✅ Milestone: Auth system live.

Phase 2: Hotels & Bookings (Weeks 4–6)

Week 4:

- Implement `Hotel` entity + endpoints.
- Insert 5–10 fake hotels in DB.

Week 5:

- Build HotelCard + Home page.
- Build Hotel Detail page.

Week 6:

- Implement `Booking` entity + endpoints.
- Add booking form to Hotel Detail page.
- Create "My Bookings" page.
  ✅ Milestone: End-to-end booking flow.

Phase 3: Blog + Polish + Deployment (Weeks 7–10)

Week 7:

- Build Blog entity + CRUD endpoints.
- Build Blog pages (list, create, edit, delete).

Week 8:

- Add UI polish:
  - Loading spinners.
  - Error messages.
  - Success toasts.
- Make responsive with Tailwind.

Week 9:

- Write clean **README.md**:
  - Overview + screenshots.
  - Tech stack.
  - Setup instructions.
  - Live demo link placeholder.

Week 10:

- Deploy backend (Render/Fly.io).
- Deploy frontend (Netlify/Vercel).
- Update README with live links.
  ✅ Final Milestone: Working, deployed app.

📌 Extra (Optional if time permits)

- Add search/filter for hotels.
- Add pagination on blogs.
- Add image upload (S3/Cloudinary).
- Write unit tests (JUnit for backend, React Testing Library).